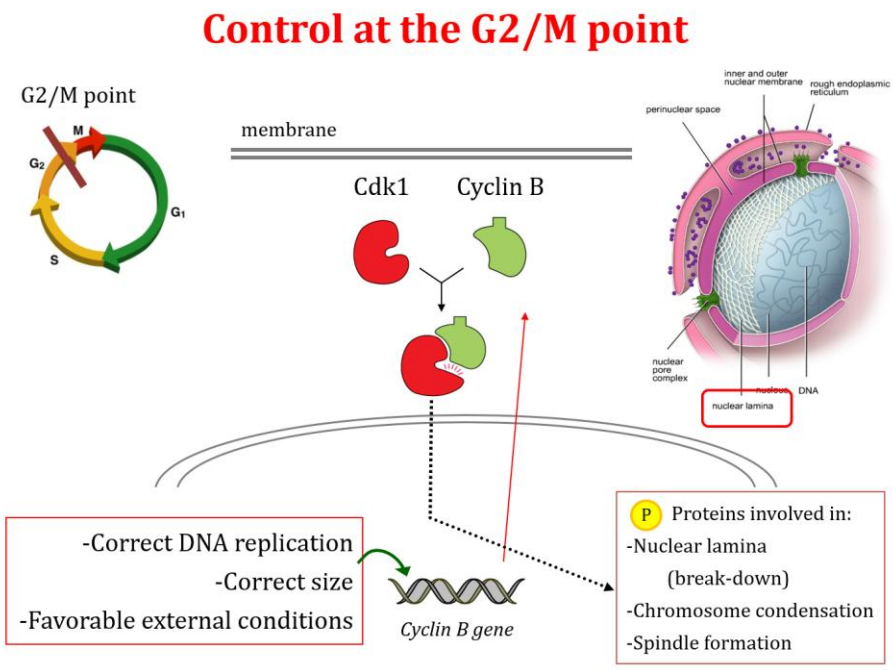
Cell Cycle Control at the G2/M Checkpoint and Mechanisms That Stop Cell Cycle Progression

G2/M Checkpoint

The G2/M checkpoint is very similar to the G1/S transition, but in this case, it controls the transition from G2 to M phase. What happens at this checkpoint? **CDK1** is the key protein responsible for controlling this transition. Cells assess whether DNA replication has been completed correctly and whether the cell has reached an adequate size. They also check whether external conditions are appropriate for cell division. If the entire genome has been correctly replicated, the cell is large enough to divide, and external conditions are favorable, the cell enters M phase. Cell size is important because the mother cell is going to produce two daughter cells, each of which needs to contain enough cytoplasm and cellular components to survive and function properly.

If the cellular sensors verify that all the conditions are appropriate, they promote the production of **cyclin B**, which is the regulatory partner of CDK1. The binding of CDK1 to cyclin B forms an active **CDK1–cyclin B complex**. This complex translocates into the nucleus and triggers several key downstream events:



A) Phosphorylation of the nuclear lamina

The nuclear lamina is a meshwork that supports the nuclear envelope and provides a structural framework for the nucleus. It also contributes to the organization and attachment of chromatin. The proteins that constitute the nuclear lamina are **lamins**, which are intermediate filament proteins. What is the consequence of phosphorylation of the nuclear lamina? **The nuclear lamina breaks down**, causing the nuclear envelope to disassemble. As a result, the chromosomes become accessible to the mitotic spindle and can be properly aligned during mitosis.

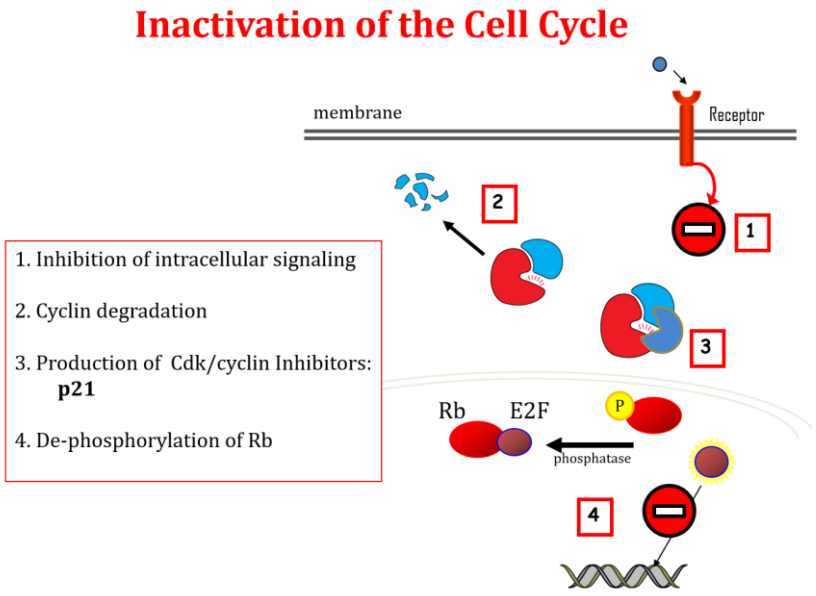
B) Chromosome condensation and spindle formation

The active **CDK1–cyclin B complex** also phosphorylates proteins that are directly involved in **chromosome condensation**, allowing the DNA to become highly compacted into visible

chromosomes. In addition, it phosphorylates proteins involved in the **formation and organization of the mitotic spindle**.

Mechanisms That Stop Progression Through the Cell Cycle

An important message is that progression through the cell cycle and mitosis is essential for tissues because cells need to divide and replace other cells that have died or become damaged. However, **stopping cell-cycle progression is equally important**. If cell division is not properly controlled, cells may proliferate uncontrollably, potentially leading to tumor formation. What mechanisms do cells use to stop the cell cycle at different stages?



1. Inhibition of signaling pathways that initiate the G1/S transition

The first mechanism is to stop the signaling cascades that initiate the transition from G1 to S phase. For example, extracellular molecules can bind to their receptors and initiate a signaling cascade that ultimately activates the expression of cyclins. Once the signal is no longer required, other proteins can inhibit this signaling pathway. This provides a mechanism for stopping signals originating outside the cell and preventing unnecessary cell-cycle progression.

2. Degradation of cyclins

Another mechanism for stopping cell-cycle progression, which we have already discussed, is the **degradation of cyclins**. Cyclins are produced at specific stages of the cell cycle, and cellular mechanisms subsequently target them for degradation. This ensures that CDK activity is tightly regulated. If cyclins were not degraded, CDKs could remain active, potentially causing cells to continue proliferating in an uncontrolled manner.

3. Inhibition of CDK–cyclin complexes

Cells can also stop cell-cycle progression through the action of **CDK–cyclin complex inhibitors**. One important protein to know is **p21**. The name p21 refers to its approximate molecular weight. If a problem occurs during the cell cycle, such as **DNA damage**, cell-cycle progression must be stopped so that the cell has time to repair the damage. In response to DNA damage, proteins such as p21 can be expressed and bind to CDK–cyclin complexes, inhibiting their activity until

the damage has been repaired. This is important because, if damaged DNA were replicated and passed on to daughter cells, genetic defects could be transmitted to the progeny.

Alterations in genes involved in cell-cycle regulation are frequently found in tumors, such as mutations in the gene coding for p21. Loss of functional cell-cycle inhibitors can allow cells to continue dividing despite the presence of DNA damage. Although the accumulation of genetic defects can ultimately lead to cell death, tumor cells can sometimes acquire additional alterations that allow them to survive and continue proliferating despite these defects. Another important example is the Rb (retinoblastoma) pathway. If the RB1 gene is altered and functional Rb protein is lost, E2F transcription factors are no longer properly inhibited. As a result, E2F can continuously activate the expression of genes required for cell-cycle progression, promoting uncontrolled proliferation.

4. Inactivation of transcription factors and intracellular signaling pathways

Cells can stop cell-cycle progression by inhibiting transcription factors and intracellular signaling pathways that promote the production of cyclins. One example involves the regulation of the **Rb protein by phosphorylation and dephosphorylation**. During G1 progression, **CDK4–cyclin D** phosphorylates Rb. This reduces Rb-mediated inhibition of **E2F**, allowing E2F to bind to the promoter regions of genes involved in DNA synthesis and cell-cycle progression. Once the appropriate signaling has occurred, **phosphatases** can remove phosphate groups from Rb. Rb then returns to its original state and can once again inhibit E2F. This helps prevent excessive or inappropriate cell proliferation.